Random-walk level noise breaks anytime-valid trend monitoring: a failure theorem, a universal e-process and the price of robustness

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Abstract

Anytime-valid monitoring of a trend usually assumes independent noise around a smooth path. This fails under the smallest departure: if the latent level carries a random-walk disturbance of any positive variance, the variance of every fixed-shape contrast statistic is inflated in proportion to the square of the series length, and e-processes built on independent noise declare a change with probability tending to one. We construct a universal e-process that is valid for any parametric Gaussian covariance class with unknown scale and a linear trend, by reducing the data to their maximal invariant under location and scale and dividing a mixture over alternatives by the supremum of the null likelihood. An information ceiling shows that no valid monitor can have expected log e-value or power above explicit values, and the e-process falls short of it by roughly the logarithm of the size of its mixture. In 56 waves of the U.S. Census Bureau's Business Trends and Outlook Survey on business use of artificial intelligence, roughness is highly significant, the independence-based monitor declares changes in nearly every series simulated from the fitted null, and the universal e-process reaches an e-value of 18 against a threshold of 20.

Keywords: anytime-valid inference; e-process; change detection; random-walk noise; survey time series

1 Introduction

Users of a sequence of published estimates inspect each release and ask whether the growth of the indicator has changed. Testing at every release inflates error rates (Armitage et al., 1969), and the modern remedy is anytime-valid inference, in which e-values and test martingales control the error uniformly over time (Howard et al., 2021; Vovk and Wang, 2021; Ramdas et al., 2023; Grünwald et al., 2024). Constructions for change detection and sequential experiments take the observations to be, given the past, draws from a fixed parametric family with independent noise (Johari et al., 2022; Shin et al., 2023, 2024). For a smooth trend this means independent errors around a deterministic path, and the e-process monitors departures of the path.

The problem. Independent noise around a straight line is rarely credible for a published series. The latent level moves from wave to wave for reasons unrelated to growth, and a random walk with small variance is the simplest description (Harvey, 1990; Harvey and Chung, 2000; Durbin and Koopman, 2012). In the fixed-sample trend-testing literature a persistent noise component is known to invalidate tests that assume independence, and it is the reason for spurious regressions (Granger and Newbold, 1974; Phillips, 1986) and for the robust trend tests of Vogelsang (1998),

Bunzel and Vogelsang (2005), Harvey et al. (2007) and Perron and Yabu (2009). Those results do not transfer to the sequential setting. A fixed-sample test is wrong at one sample size; a monitor that is inspected at every release and uses a threshold that grows only logarithmically is wrong in a way that worsens with every release. Nothing in the anytime-valid literature quantifies this, and none of its constructions has a null class that contains a rough level.

Contributions.

1. *A failure theorem* (Section 3). For any fixed-shape contrast of an equally spaced series, a random-walk level of variance p per wave and sampling variance σ^2 inflate the variance of the standardized statistic to $1 + (p/\sigma^2)\, c\, n^2\, (1 + o(1))$ with $c > 0$ depending on the contrast only (for curvature $c = 1/42$). The independence-based e-process therefore declares with probability tending to one for every $p > 0$. At the 54 waves of our application the inflation is 150-fold, and the failure begins when $n \approx \sigma/\sqrt{pc}$, about five waves.

2. *A universal e-process and its validity theorem* (Section 4). The data are reduced to the maximal invariant under the location nuisance and the unknown scale, the numerator is an arbitrary mixture over alternatives, and the denominator is the supremum of the invariant null likelihood over a correlated parametric class. The e-process is valid for every member of the class (Theorem 2); the random-walk level, an autoregressive level and a random slope are special cases. The reduction by scale invariance is exact and removes the scale from both the denominator and the mixture; in simulation it raises power from 0.39 to 0.70 for a slope change of 0.4 points per wave.

3. *An information ceiling* (Section 4.3). For any monitor that is valid for the null class we give a closed-form upper bound on expected log e-value (the smallest Kullback–Leibler divergence from the alternative to the null class) and on power. The ceiling is a lower bound on what any monitor can detect. On the design of our application no valid monitor can detect a slope change of 0.09 points per wave with probability 50 percent, against an average growth of 0.12, and the e-process reaches 50 percent at 0.35. The gap between the ceiling and the achieved growth of the log e-value is of the order of the log of the mixture size.

4. *An application* (Section 6). On the 56 waves of the Business Trends and Outlook Survey (BTOS) on business use of artificial intelligence, roughness is highly significant (likelihood-ratio statistic 79.6), an independence-based monitor declares two or more changes in 98 percent of series simulated from the fitted null, and the universal e-process reaches an e-value of 18 against a threshold of 20. We report this as borderline and not as a detection.

Related work. The e-process machinery is standard: Ville's inequality, mixture martingales (Robbins, 1970), invariance with respect to a nuisance group (Pérez-Ortiz et al., 2024) and the composite null of universal inference (Wasserman et al., 2020; Ruf et al., 2022). Sequential change detection is typically built on cumulative-sum or generalized likelihood-ratio statistics with calibrated thresholds (Page, 1954; Siegmund and Venkatraman, 1995; Lorden and Pollak, 2005); e-detectors (Shin et al., 2024) give anytime-valid versions for independent observations. We propose no new mixture or detector. The contributions are the failure result, which has no counterpart in the anytime-valid literature; the validity theorem for a correlated null class with exact scale

invariance; the information ceiling, which separates what is lost to the null class from what is lost to the construction; and the application. State-space models for survey series with a rough level (Scott and Smith, 1974; Durbin and Koopman, 2012; Harvey and Chung, 2000) estimate the trend and level but give no sequential error control, and their likelihood-ratio tests for roughness have nonstandard null distributions (Self and Liang, 1987).

2 Setting

Let $y_1, \dots, y_n$ be published estimates at release times $t_1 < \cdots < t_n$ (in units of the release interval) with published design-based standard errors s_j, and let ρ_j label the regime (here, the questionnaire wording). Write $T_j = t_j - t_1$, $D = \mathrm{diag}(s_1^2, \dots, s_n^2)$ and $W_{jk} = \min(T_j, T_k)$. The null class is

$$y \sim \mathcal{N}\big(Xc,\ \kappa^2(D + r\,W)\big), \qquad c \in \mathbb{R}^q,\ \kappa^2 > 0,\ r \in \mathcal{R}_0, \tag{1}$$

where X has one column per regime (an unknown level for each) and one column t_j (an unknown common slope). Equation (1) is the marginal law of $y_j = L_j + e_j$ with sampling error $e_j \sim \mathcal{N}(0, \kappa^2 s_j^2)$ and level $L_j = L_1 + b\,T_j + \sum_{i \le j} \eta_i$, $\eta_i \sim \mathcal{N}(0, \kappa^2 r\,(T_i - T_{i-1}))$: a linear trend plus a random walk. The roughness $r \ge 0$ is measured relative to κ^2, and $p = \kappa^2 r$ is the variance of the level disturbance per wave. The roughness is a statistical description of the departures of the level from a line; we do not attribute it to a cause, and it is not identified against sampling error that is correlated across waves (for example through panel overlap), which the published standard errors, treating waves as independent, would understate. The set $\mathcal{R}_0$ is a finite grid that contains zero; we use 46 values spanning four orders of magnitude. The covariance of the first n waves is the leading block of that of the first $n+1$, so (1) defines a family of laws on the sequence.

Two families of alternatives are monitored, each at an unknown wave k and with an unknown size δ: a change of slope, $\mathbb{E}y = Xc + \delta\,(t - t_k)_+$, and a level shift, $\mathbb{E}y = Xc + \delta\,\mathbf{1}\{j \ge k\}$. A level shift of this kind is what an unannounced instrument event looks like. Both the horizon $n_{\max}$ and all grids and prior scales used below are fixed before the data are seen; the application is retrospective.

3 Independence-based monitors fail under level roughness

A contrast statistic. Take an equally spaced series, $t_j = j$, with $s_j = \sigma$. Let $a \in \mathbb{R}^n$ be a contrast orthogonal to the null design, $X^\top a = 0$ (for this section, $X = (1, j)$), and

$$Z_a = \frac{a^\top y}{\sigma \|a\|},$$

which is $\mathcal{N}(0,1)$ under independence. Curvature, a slope change at a fixed wave and a level shift at a fixed wave are all of this form after the null design is projected out. The independence-based monitor with prior scale τ declares when the mixture e-process $E_n(a) = (1 + \tau^2 B)^{-1/2} \exp\{\tau^2 A^2 / (2(1 + \tau^2 B))\}$, with $A = a^\top y / \sigma^2$ and $B = \|a\|^2 / \sigma^2$, reaches $2/\alpha$ (Robbins, 1970); the variant with a half-normal prior behaves identically for our purposes and is the one used in the simulations.

Theorem 1 (failure)**.** *Let $y_j = \beta_0 + \beta_1 j + L_j + e_j$ with $e_j \sim \mathcal{N}(0, \sigma^2)$ independent and L a Gaussian random walk with increment variance $p > 0$. Let $a = a^{(n)}$ be the residual of $\psi(j/n)$ on $(1, j)$, where*

Table 1: Inflation of independence-based contrasts on the BTOS design (54 legacy-wording waves, fitted roughness $\hat{r} = 0.045$, scale $\hat{\kappa}^2 = 1.28$). λ: Rayleigh quotient $r\,a^\top W a / a^\top D a$; Var of Z: variance of the contrast standardized with the published standard errors, which equals $\hat{\kappa}^2(1+\lambda)$; the last column is the probability that a fixed-n test of nominal level 5% rejects.

Contrast	Inflation λ	Var of Z	False positives of a nominal 5% test (%)
curvature	151	194	89
slope change at wave 20	156	201	89
level shift at wave 20	72	94	84
slope change at wave 35	135	175	88
level shift at wave 35	97	125	86

ψ is piecewise continuous on $[0,1]$ and is not almost everywhere equal to a linear function. Put $\tilde{\psi}$ for the residual of ψ on $(1,x)$ in $L_2[0,1]$ and $\Psi(x) = \int_x^1 \tilde{\psi}$. Then

$$\mathrm{Var}(Z_a) = 1 + \lambda_n, \qquad \lambda_n = \frac{p}{\sigma^2}\,\frac{a^\top W a}{\|a\|^2} = \frac{p}{\sigma^2}\,c(\psi)\,n^2\,(1+o(1)), \qquad c(\psi) = \frac{\int_0^1 \Psi^2}{\int_0^1 \tilde{\psi}^2} > 0.$$

Consequently, for every $\tau > 0$, $\alpha \in (0,1)$ and $p > 0$, the probability that $E_n(a) \geq 2/\alpha$ tends to one as $n \to \infty$. For curvature, $\psi(x) = x^2$ and $c = 1/42$.

The proof is in Appendix A. The quantity λ_n is the Rayleigh quotient of W against the sampling covariance on the contrast, and it is exact for any a; the asymptotic form gives the rate. Three consequences follow. First, the phase transition: the inflation exceeds one when $n > n^* = \sigma/\sqrt{p\,c}$; for the curvature of the BTOS design, with $r/s^2 \approx 2$, this is about five waves. Second, the statement is not an artifact of the half-normal mixture or of known scale; Proposition 1 treats estimation of the scale. Third, the failure is in the variance, not in a bias: the statistic is centered and a simple recalibration of its threshold would restore the level at one n but not uniformly over n, since the inflation grows.

Proposition 1 (unknown scale; proof sketch)**.** *For curvature, the same conclusion holds for the version of the e-process in which σ^2 is estimated from the residual sum of squares of the fit of $(1,j)$: the studentized statistic diverges at rate $\sqrt{n}$.*

Table 1 shows what this means for the application design: contrasts with the variance inflated 94-fold to 201-fold, and fixed-sample tests that reject in 84 to 89 percent of series simulated from a straight line. Figure 1 shows the anytime-valid case. With $p = 0.01$, under half the sampling variance $\sigma^2 = 0.0225$, the false-declaration probability of the monitor is 0.31 at 20 waves, 0.76 at 40, 0.97 at 80 and 1.0 at 160.

4 A universal e-process

4.1 Construction and validity

Let X_n be the design of the first n waves and $R_n = M_n y_{1:n}$ the data modulo the column space of X_n, with M_n the orthogonal projection onto its complement and $\nu = n - q$. Under (1) the law of R_n does not depend on c. The scale κ^2 is a further nuisance: multiplying y by a constant multiplies

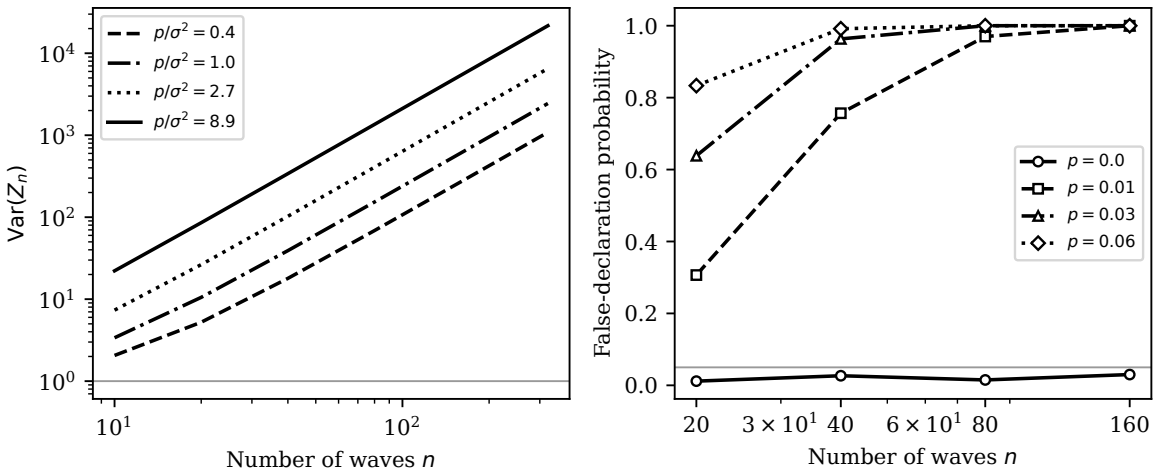


Figure 1: The independence-based e-process under a rough level. Left: probability of having declared a curvature change by wave n (known scale, threshold 40 on each of two one-sided monitors, level 0.05) for several p; sampling standard deviation 0.15. Right: the empirical variance of the standardized statistic against the theoretical $1 + (p/\sigma^2)n^2/42$.

R_n by the same constant. The maximal invariant under both is the direction $u_n = R_n/|R_n|$. Under $\mathcal{N}(Xc, \kappa^2\Sigma_r)$, with $\Sigma_r = D + rW$, the direction has an angular central Gaussian law that depends on r only, with density, relative to the uniform measure on the sphere of the residual space,

$$p_r(u_n) \propto |\Sigma_r|^{-1/2}\,|X_n^\top \Sigma_r^{-1} X_n|^{-1/2}\,\big(y^\top P_r\, y\big)^{-\nu/2}, \qquad P_r = \Sigma_r^{-1} - \Sigma_r^{-1} X_n (X_n^\top \Sigma_r^{-1} X_n)^{-1} X_n^\top \Sigma_r^{-1},$$

up to a factor $|R_n|^\nu$ and a constant that are the same for all r and all alternatives and cancel in every ratio below.

A component $c = (r_g, f, k, \tau)$ of the alternative has a roughness r_g, a family f (slope or level), a change wave k and a prior scale τ. With z the alternative column, $a = z^\top P_{r_g} y$, $b = z^\top P_{r_g} z$ and $S = y^\top P_{r_g} y$, put

$$\frac{q_c(u_n)}{p_{r_g}(u_n)} = (1+\tau^2 b)^{-1/2}\Big(1 - \frac{\tau^2 a^2}{(1+\tau^2 b)\,S}\Big)^{-\nu/2}. \tag{2}$$

This is the ratio of the laws of u_n when $y \sim \mathcal{N}(Xc + \delta z, \kappa^2\Sigma_{r_g})$ with $\delta \mid \kappa \sim \mathcal{N}(0, \kappa^2\tau^2)$ and when $\delta = 0$: integrating out δ gives covariance $\kappa^2(\Sigma_{r_g} + \tau^2 z z^\top)$, again an angular central Gaussian, and the determinant and quadratic form follow from the matrix determinant lemma and the Sherman–Morrison formula. The expression in parentheses is positive because $a^2 \le bS$. Let $Q_n = \sum_c w_c q_c$ with fixed weights, and define

$$E_n = \frac{Q_n(u_n)}{\max_{r\in\mathcal{R}_0} p_r(u_n)}. \tag{3}$$

The weights are uniform over two families, change waves $k \in \{3, \dots, n_{\max}\}$, five roughness values from zero to 32 times the median sampling variance, and four prior scales per family (slope: 0.1, 0.2, 0.4, 0.8; level: 0.5, 1, 2, 4), about 2,100 components for $n_{\max} = 54$ and a log-weight of 7.6 nats. Components whose change wave lies beyond the current wave have $z = 0$ on the data so far and contribute the null density. The supremum in (3) is over the 46 roughness values of $\mathcal{R}_0$ and nothing

is profiled: the scale has been removed by invariance, not estimated.

Theorem 2 (validity)**.** *Let $\{\Sigma_\theta(n)\}_{\theta\in\Theta_0}$ be any collection of positive-definite covariance matrices, consistent in n (each $\Sigma_\theta(n)$ is the leading block of $\Sigma_\theta(n+1)$), and X_n designs for which the column space of X_{n+1} restricted to the first n rows is that of X_n. Let Q_n be any mixture of consistent angular Gaussian laws of u_n, and let E_n be $Q_n(u_n)/\sup_{\theta\in\Theta_0} p_\theta(u_n)$. For every $\theta\in\Theta_0$, every c and every $\kappa^2>0$, if $y\sim\mathcal{N}(Xc,\kappa^2\Sigma_\theta)$ then $\mathbb{P}(\exists n: E_n\geq 1/\alpha)\leq\alpha$. The same holds with the Gaussian laws of R_n in place of u_n, a null class in which κ^2 is a parameter of Θ_0, and the supremum taken over it (the profile-scale variant).*

Corollary 1. *Theorem 2 covers the random-walk level of* (1)*, a first-order autoregressive level with unknown correlation, a random slope (integrated random walk), and any finite or countable union of such classes.*

The proof (Appendix A) shows that $\Lambda_n = Q_n/p_\theta$ is a likelihood-ratio martingale on the nested σ-fields generated by u_n and that $E_n\leq\Lambda_n$ because the supremum is at least p_θ; Ville's inequality then gives the claim. The point that needs care is that u_n is a function of u_{n+1}, which holds because normalizing cancels the scale, so that the direction of a coordinate projection is a function of the direction of the full vector. Nothing depends on how Q_n is mixed, so the construction is a template. The profile-scale variant, which estimates κ^2 by its restricted maximum-likelihood value in the denominator and mixes over four values of it in the numerator, is also valid and is the one we used in an earlier analysis; the exact-scale version has fewer mixture components, no profiling, and more power (Section 5).

4.2 Cost decomposition

Proposition 2 (decomposition)**.** *For every component c with roughness $r_g\in\Theta_0$,*

$$\log E_n \;\geq\; \underbrace{\log w_c}_{\text{mixture}} + \underbrace{\log\Lambda_{c,n}}_{\text{known-}r\text{ LR}} - \underbrace{\Delta_n}_{\text{supremum}}, \qquad \Delta_n = \log\max_{r\in\mathcal{R}_0} p_r(u_n) - \log p_{r_g}(u_n) \;\geq 0, \tag{4}$$

where $\Lambda_{c,n} = q_c/p_{r_g}$ is the likelihood ratio of a user who knows the roughness and the component.

The mixture costs $\log(1/w_c)$, 7.6 nats here. The supremum costs Δ_n, which under the null at r_g is a likelihood-ratio statistic for one parameter and is $O_P(1)$; under an alternative the null fit can absorb signal, and Δ_n is measured and not bounded (Section 5).

4.3 An information ceiling

How much of the loss is due to the construction, and how much is due to the null class? The answer does not depend on any particular e-process.

Theorem 3 (ceiling)**.** *Fix n, a null class $\{P_\theta\}_{\theta\in\Theta_0}$ of laws of the data and an alternative Q, and let $K=\inf_{\theta\in\Theta_0}\mathrm{KL}(Q\,\|\,P_\theta)$.*

(i) If $E\geq 0$ satisfies $\mathbb{E}_\theta E\leq 1$ for all $\theta\in\Theta_0$, then $\mathbb{E}_Q\log E\leq K$.

(ii) If A is an event with $P_\theta(A)\leq\alpha$ for all $\theta\in\Theta_0$, then $Q(A)\leq\beta^*(K,\alpha)=\max\{\beta:\mathrm{kl}(\beta\,\|\,\alpha)\leq K\}$, with kl *the binary relative entropy.*

(iii) Let the null class be $\mathcal{N}(0, \kappa^2 M\Sigma_r M^\top)$ *on the residual space,* $\kappa^2 > 0$, $r \in \mathcal{R}_0$, *and* $Q = \mathcal{N}(\delta M z, \kappa_g^2 M\Sigma_{r_g} M^\top)$. *Then*

$$K = \min_{r\in\mathcal{R}_0} \tfrac{1}{2}\Big[\nu \log\frac{c_r}{\nu} + \log|M\Sigma_r M^\top| - \log|\kappa_g^2 M\Sigma_{r_g} M^\top|\Big], \quad c_r = \operatorname{tr}\big((M\Sigma_r M^\top)^+ \kappa_g^2 M\Sigma_{r_g} M^\top\big) + \delta^2 z^\top P_r z,$$

and $K \le \frac{1}{2}\nu\log(1 + \delta^2 b_g/\nu) \le \frac{1}{2}\delta^2 b_g$, $b_g = z^\top P_{r_g} z/\kappa_g^2$.

Both bounds hold for the monitor $\{E_n\}$ *at any horizon, since* $\{\exists n \le n_{\max} : E_n \ge 1/\alpha\}$ *is an event of probability at most* α *under every* P_θ, *and for the scale-exact monitor, whose data are a function of the full data and whose divergence is therefore no larger.*

Part (iii) shows what the null class does: the roughness (and the scale) can absorb part of the signal, so that K is strictly smaller than the information $\frac{1}{2}\delta^2 b_g$ that a user who knew the null would have. The ceiling is on the design alone and does not depend on the construction: it applies to every valid monitor, including those that are tuned to the alternative. We use it in two ways. In Section 5 it is compared with the growth of the log e-value that the universal e-process achieves. In Section 6 it gives a lower bound on the minimum detectable change.

5 Simulation

All simulations use the dates and published standard errors of the 54 legacy-wording BTOS waves, a linear mean of slope 0.12 points per wave, level $\alpha = 0.05$ (threshold 20 for a single process), $n_{\max} = 54$ and the exact-scale e-process unless stated.

Validity. The probability that the e-process ever reaches 20 is 0.000 for independent errors, 0.000 for standard errors 1.5 times the published ones, and 0.002, 0.008 and 0.003 for random-walk levels with $p = 0.03$, 0.06 and 0.2 per wave (600 replications each), against 1.00 for the independence-based monitor at $p = 0.06$. For autoregressive levels with correlation 0.9 and 0.98, which lie outside the random-walk class, the profile-scale version was also never above the threshold, and enlarging its class with an autoregressive-level setting changed the power at $p = 0.06$, slope 0.4 from 0.41 to 0.39. The realized probabilities are far below α: the construction is conservative.

Power. Table 2 compares the exact-scale and profile-scale versions. The exact-scale version has uniformly higher power in the cases that matter: at $p = 0.06$ a slope change of 0.4 points per wave is detected in 70 percent of replications against 43 percent, and a level shift of 2 points in 88 against 76 percent. Without roughness both detect a slope change of 0.2 in nearly all replications; with $p = 0.06$ the same change is detected in 7 and 1 percent. Roughness removes most of the power of any monitor, which Section 6 quantifies by the ceiling.

Where the cost comes from. Table 3 separates the terms of (4). A user who knows the roughness, the family, the size and the wave detects a slope change of 0.4 at $p = 0.06$ in 99.8 percent of replications. Adding the supremum over the null roughness, for a single component, reduces this to 98 percent. The mixture over about 2,100 components reduces it to 70 percent. With exact scale invariance the supremum is nearly free; the cost is the mixture, the breadth of the

Table 2: Probability of declaring by the last wave when the slope changes by δ points per wave, or the level shifts by δ points, at wave 25 of 54, for roughness p (variance of the level disturbance per wave; the published sampling variance is about 0.02). 300 replications.

p	Alternative	Profile scale	Exact scale
0.0	slope change 0.2	0.98	1.00
0.03	slope change 0.2	0.08	0.17
0.06	slope change 0.2	0.01	0.07
0.06	slope change 0.4	0.43	0.70
0.06	level shift 1.0	0.02	0.07
0.06	level shift 2.0	0.76	0.88
0.03	level shift 1.0	0.12	0.22

Table 3: Probability of declaring by the last wave at level 0.05 for the known-roughness likelihood ratio, a single component with the supremum denominator, and the full universal e-process (400 replications, change at wave 25).

p	Alternative	Known-r LR	One component, sup	Universal
0.06	slope change 0.4	1.00	0.98	0.69
0.06	slope change 0.2	0.50	0.35	0.05
0.06	level shift 2.0	0.99	0.99	0.90
0.06	level shift 1.0	0.67	0.61	0.11
0.03	slope change 0.2	0.86	0.67	0.15
0.03	level shift 1.0	0.86	0.83	0.20

alternatives. At the smaller changes the supremum matters more (0.50 to 0.35 for a slope change of 0.2), and the mixture takes most of the rest.

Comparison with a calibrated test. A user who knew the null class could calibrate a likelihood-ratio monitor by simulation. We took the maximum over waves, families and change waves of the generalized likelihood ratio with maximum-likelihood nuisance parameters, and set the threshold at the 95 percent quantile of its null distribution at the least favorable roughness. Table 4 shows that this calibrated monitor is no more powerful than the profile-scale e-process in the cases that matter and is clearly less powerful than the exact-scale version (0.23 against 0.70 for slope 0.4 at $p = 0.06$; 0.70 against 0.88 for a level shift of 2). The e-process gives anytime validity with no calibration.

Growth against the ceiling. Table 5 compares the mean log e-value at the last wave with the ceiling K of Theorem 3, for alternatives placed at wave 25 of the fitted 56-wave BTOS design (400 replications). The ceiling holds in every case, as it must. The gap grows from about 4 nats at the smallest changes to 9 at the largest, and is of the order of the log-weight of the mixture, 7.6 nats, plus the cost of the supremum. The universal e-process therefore loses to the best possible monitor a quantity that is a constant in n, the price of not knowing the alternative, and not a loss of rate.

Table 4: Probability of declaring for the calibrated generalized likelihood-ratio monitor and for the two e-processes (300–400 replications; independent replications, so differences of 0.05 are within noise).

p	Alternative	Calibrated GLR	Profile-scale e-process	Exact-scale e-process
0.0	slope change 0.2	1.00	0.99	1.00
0.03	slope change 0.2	0.10	0.03	0.17
0.06	slope change 0.2	0.04	0.01	0.07
0.06	slope change 0.4	0.23	0.39	0.70
0.0	level shift 1.0	0.99	0.96	1.00
0.03	level shift 1.0	0.12	0.10	0.22
0.06	level shift 1.0	0.07	0.03	0.07
0.06	level shift 2.0	0.70	0.78	0.88

Table 5: Mean log e-value at wave 56 of the exact-scale e-process against the ceiling K of Theorem 3, on the fitted BTOS design with change at wave 25 (nats).

Alternative	Mean $\log E_{56}$	Ceiling K	Gap
slope change 0.2	-0.2	3.5	3.7
slope change 0.3	1.5	6.6	5.1
slope change 0.4	3.9	9.9	6.0
slope change 0.5	6.7	13.4	6.7
slope change 0.6	10.0	17.1	7.2
level shift 1.0	-0.6	4.6	5.3
level shift 1.5	2.3	9.3	7.1
level shift 2.0	6.5	14.6	8.2
level shift 2.5	11.9	20.0	8.1
level shift 3.0	16.6	25.2	8.7

6 Application: business use of artificial intelligence

Data. The BTOS is the Census Bureau's biweekly survey of about 1.2 million employer businesses (U.S. Census Bureau, 2025). We use the 56 national waves on current use of artificial intelligence from 24 September 2023 to 14 December 2025 (Bonney et al., 2024): 54 with the legacy wording and two with the wording introduced on 17 November 2025, which has its own level. The estimate rises from 3.7 to 10.0 percent under the legacy wording; the published standard errors have median 0.15.

Roughness and model choice. Maximizing the restricted likelihood over (1) gives $\hat{p} = 0.058$ points^2 per wave (95 percent profile interval 0.018 to 0.46) and $\hat{\kappa}^2 = 1.28$; the level disturbance has standard deviation 0.24 points per wave against a sampling standard deviation of 0.15. Setting $r = 0$ lowers the restricted log-likelihood by 39.8, a likelihood-ratio statistic of 79.6; because $r = 0$ is on the boundary, the reference distribution is the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ (Self and Liang, 1987) and the evidence is overwhelming. Table 6 compares models by AIC. A straight line with a random-walk level is best; adding curvature does not improve it, and independent noise is far worse even with a quadratic mean.

Table 6: Maximum-likelihood AIC of competing models for the legacy-wording national series on current use and on planned use of artificial intelligence (lower is better; regime levels are in all models).

Model (regime levels in all)	AIC, current use	AIC, planned use
line + independent noise	114.5	145.6
quadratic + independent noise	70.9	62.2
line + random-walk level	40.9	45.3
quadratic + random-walk level	41.9	42.0
line + random slope (integrated random walk)	52.5	53.9
line + random-walk level + random slope	42.9	47.1
line + AR(1) level (phi=0.9)	42.2	49.8

What an independence-based monitor would say. The monitor of Section 3, restarted after each declaration, declares twice on the BTOS series, on 10 March 2024 (deceleration) and 6 April 2025 (acceleration). In 400 series simulated from the fitted null (straight line, estimated roughness and scale, BTOS dates and standard errors) it declares at least once in 100 percent and at least twice in 98 percent, 3.1 times on average. The two declarations are what the fitted null produces.

The universal e-process. Applied to the 56 waves, the exact-scale e-process peaks at $\log E = 2.88$ ($E = 17.8$) on 27 July 2025, against a threshold of $\log 20 = 3.00$ (Figure 2); it does not declare. The evidence comes from the level family ($\log E = 3.56$ on its own, with the slope family at -0.08); the combined e-process, which weights the two families equally, pays $\log 2$ for the choice. We do not interpret it as an event. Table 7 shows that the conclusion is sensitive to the design of the mixture: restricting the nuisance mixture to rough settings or widening the prior scales brings $\log E$ to 3.30 and 2.99, and level alternatives alone give 3.56. These variants were run after the main analysis and we do not take them as detections; they show that the main result is a near miss that depends on the dilution of the mixture, not a clear absence of evidence. The profile-scale version of the e-process, which we applied first, peaks at 0.39. The seven employment-size classes (maximum log e-value from -0.14 to 0.08 each) and the series on planned use (0.22) show no evidence. The exact-scale e-process was developed after the profile-scale analysis, with the same grids and prior scales; the main analysis of this paper is the exact-scale one because it is the better procedure in simulation, and the sequence is disclosed here.

A positive control. The method must be able to detect a real discontinuity. When the change of questionnaire wording on 17 November 2025 is treated as ordinary data, with no regime level, the exact-scale e-process reaches $\log E = 19.2$ ($E \approx 2 \times 10^8$) and the profile-scale one 17.2. The monitor is therefore not insensitive; it does not declare changes that a rough level explains.

What the series could have revealed. We inject a change at wave 25 into series simulated from the fitted null on the full 56-wave design and record how often the e-process crosses 20 (300 replications per size), and compare with the ceiling of Theorem 3 for the same alternative (Figure 3). A slope change is detected with probability 50 percent at 0.35 points per wave and 80 percent at 0.44, and a level shift at 1.61 and 1.93 points (profile scale: 0.43, 0.51, 1.75 and 2.12). The ceiling gives a lower bound that holds for any valid monitor: a slope change cannot be detected

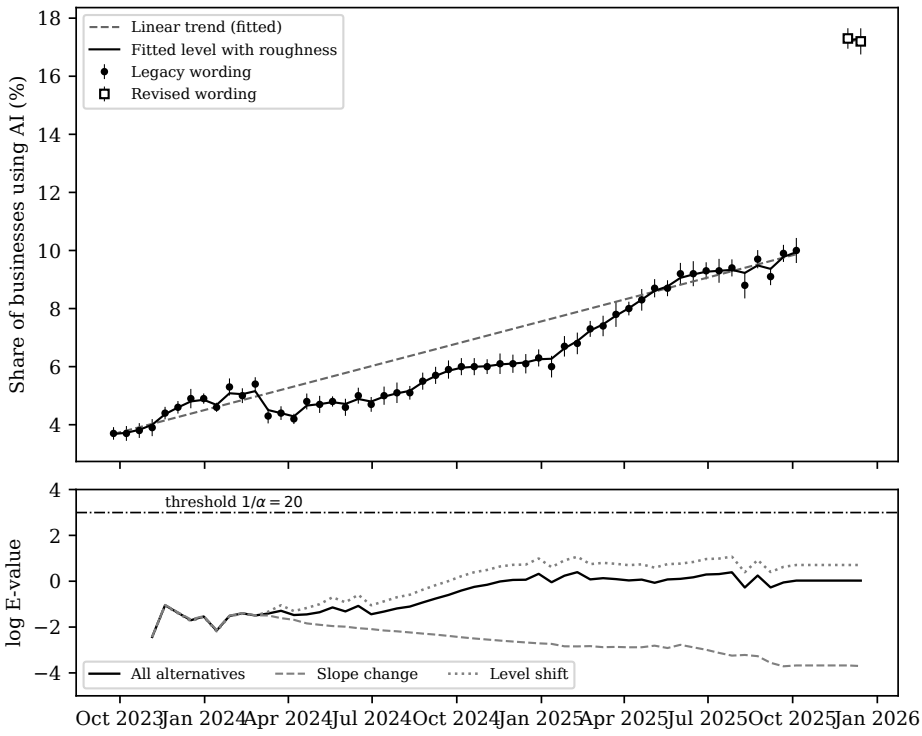


Figure 2: Top: national share of businesses using artificial intelligence with 95 percent intervals, the fitted linear trend and the fitted trend plus rough level. Bottom: log E-value of the profile-scale universal e-process, for all alternatives and by family, against the threshold $\log 20$.

with probability 50 percent below 0.09 points per wave and with probability 80 percent below 0.14, and a level shift below 0.40 and 0.63 points. The average growth is 0.12 points per wave, so even a monitor that knew the alternative would have only an even chance of detecting a change in growth equal to the growth itself. The e-process is a factor of about four above the ceiling, most of which is the price of not knowing the wave, the size and the roughness, the log-weight of the mixture in Table 5.

7 Discussion

What the paper shows. Anytime-valid monitoring of a trend is only as good as its null. If the latent level is rough, independence-based e-processes are not slightly anticonservative; they are wrong, by a margin that grows with the series, and the failure begins at roughness well below the sampling variance. A null class that includes the roughness restores validity. Exact scale invariance makes the construction more powerful than profiling the scale, and the information ceiling shows that the loss to the null class is large and unavoidable (no monitor can detect a change in growth smaller than the growth in half of the cases) while the loss to the construction is a constant of order $\log(1/w)$. On the BTOS the result is a near miss, an e-value of 18 against a threshold of 20, which the paper reports as that.

Table 7: Maximum log e-value on the BTOS national series under variants of the mixture (threshold $\log 20 = 3.00$). Variants other than the default were run after seeing the default.

Variant	Exact scale	Profile scale
Default	2.88	0.39
Wider prior scales on δ	2.99	0.10
Nuisance mixture restricted to rough settings	3.30	1.17
Slope alternatives only	−0.08	−1.03
Level alternatives only	3.56	1.07
Legacy-wording waves only	2.92	0.43

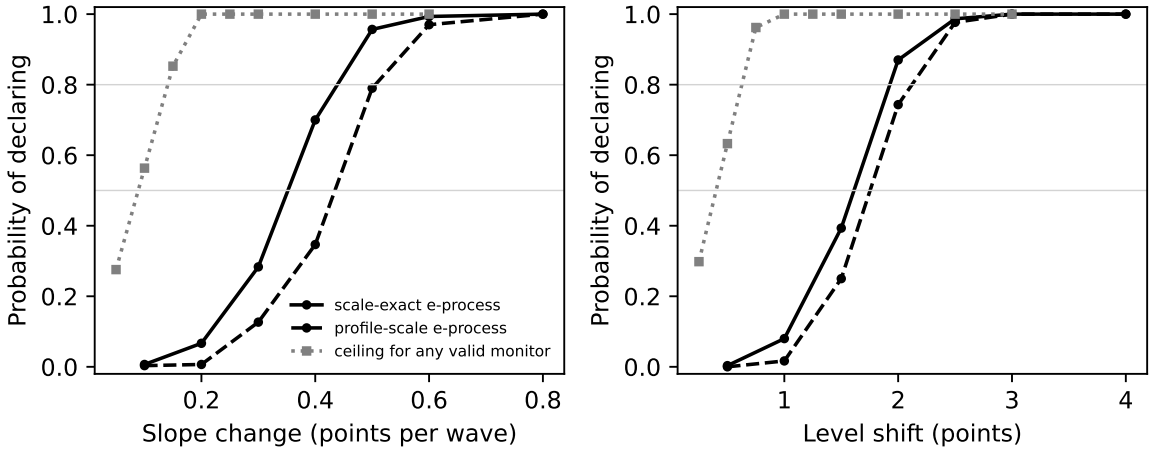


Figure 3: Probability of crossing 20 when a change is injected at wave 25 into series simulated from the fitted null on the BTOS design, for a slope change (left) and a level shift (right): exact-scale e-process, profile-scale e-process, and the ceiling for any valid monitor.

Limitations. The null class is Gaussian; the validity theorem covers any consistent parametric Gaussian covariance class but not non-Gaussian errors, and the failure theorem is for the benchmark monitor in an equally spaced design with a random-walk level. The e-process is conservative, with realized false-declaration probabilities below 0.01. The alternative mixture, the horizon and the grids are fixed in advance, and the application is retrospective; the exact-scale version was developed after the profile-scale analysis, and the sensitivity variants were run after the main analysis. The ceiling is for a known alternative and is loose for power (it ignores the cost of not knowing the alternative, which is the gap we measure), so it bounds detectability from below and not the e-process from above. The standard errors enter as fixed weights, panel rotation is not modeled, and the roughness is not identified against correlated sampling error. We analyze two series from one survey; the roughness estimates in the size classes range from 0.06 to 0.94, and a replication in another survey is needed before the magnitude of the roughness is taken as general. We prove no matching upper bound on the loss of the e-process; the gap in Table 5 is measured. A tolerance for small departures from linearity and a regret analysis are open.

Implications. A declaration from a monitor that assumes independent noise should be checked by estimating the roughness and simulating the monitor under the fitted null. For agencies, the

information ceiling and the spike-in curve are design quantities that say how large a change a series of a given length, precision and roughness can reveal, and they can be reported with the series.

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Declaration of generative AI use

Generative AI tools were used to support the writing of code and to polish the language of the manuscript. The research idea, the methods and algorithms, and the interpretation of the results are the author's own. The author reviewed and edited all AI-assisted output and takes full responsibility for the content of the work.

Data availability statement

All data are public statistics of the U.S. Census Bureau (Business Trends and Outlook Survey). The analysis data set, the code that reproduces every number, table and figure, and a Lean 4 formalization of the martingale and Ville-inequality steps of the validity proof accompany the submission.

CRediT authorship contribution statement

Quang-Vinh Dang: conceptualization, methodology, formal analysis, software, data curation, visualization, writing (original draft, review and editing).

References

Armitage, P., C. K. McPherson, and B. C. Rowe (1969). Repeated significance tests on accumulating data. *Journal of the Royal Statistical Society: Series A 132*(2), 235–244. https://doi.org/10.2307/2343787.

Bonney, K., C. Breaux, C. Buffington, E. Dinlersoz, L. Foster, N. Goldschlag, J. Haltiwanger, Z. Kroff, and K. Savage (2024). Tracking firm use of AI in real time: A snapshot from the Business Trends and Outlook Survey. NBER Working Paper 32319, National Bureau of Economic Research. https://doi.org/10.3386/w32319.

Bunzel, H. and T. J. Vogelsang (2005). Powerful trend function tests that are robust to strong serial correlation, with an application to the Prebisch–Singer hypothesis. *Journal of Business & Economic Statistics 23*, 381–394.

Durbin, J. and S. J. Koopman (2012). *Time Series Analysis by State Space Methods* (2nd ed.). Oxford: Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199641178.001.0001.

Granger, C. W. J. and P. Newbold (1974). Spurious regressions in econometrics. *Journal of Econometrics 2*, 111–120.

Grünwald, P., R. de Heide, and W. M. Koolen (2024). Safe testing. *Journal of the Royal Statistical Society: Series B 86*(5), 1091–1128. https://doi.org/10.1093/jrsssb/qkae011.

Harvey, A. C. (1990). *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9781107049994.

Harvey, A. C. and C.-H. Chung (2000). Estimating the underlying change in unemployment in the UK. *Journal of the Royal Statistical Society, Series A 163*.

Harvey, D. I., S. J. Leybourne, and A. M. R. Taylor (2007). A simple, robust and powerful test of the trend hypothesis. *Journal of Econometrics 141*, 1302–1330.

Howard, S. R., A. Ramdas, J. McAuliffe, and J. Sekhon (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics 49*(2), 1055–1080. https://doi.org/10.1214/20-AOS1991.

Johari, R., P. Koomen, L. Pekelis, and D. Walsh (2022). Always valid inference: Continuous monitoring of A/B tests. *Operations Research 70*(3), 1806–1821. https://doi.org/10.1287/opre.2021.2135.

Lorden, G. and M. Pollak (2005). Nonanticipating estimation applied to sequential analysis and changepoint detection. *Annals of Statistics 33*, 1422–1454.

Page, E. S. (1954). Continuous inspection schemes. *Biometrika 41*(1–2), 100–115. https://doi.org/10.1093/biomet/41.1-2.100.

Pérez-Ortiz, M. F., T. Lardy, R. de Heide, and P. Grünwald (2024). E-statistics, group invariance and anytime-valid testing. *Annals of Statistics 52*(4). https://doi.org/10.1214/24-AOS2394.

Perron, P. and T. Yabu (2009). Estimating deterministic trends with an integrated or stationary noise component. *Journal of Econometrics 151*, 56–69.

Phillips, P. C. B. (1986). Understanding spurious regressions in econometrics. *Journal of Econometrics 33*, 311–340.

Ramdas, A., P. Grünwald, V. Vovk, and G. Shafer (2023). Game-theoretic statistics and safe anytime-valid inference. *Statistical Science 38*(4), 576–601. https://doi.org/10.1214/23-STS894.

Robbins, H. (1970). Statistical methods related to the law of the iterated logarithm. *Annals of Mathematical Statistics 41*(5), 1397–1409. https://doi.org/10.1214/aoms/1177696786.

Ruf, J., M. Larsson, W. M. Koolen, and A. Ramdas (2022). A composite generalization of Ville's martingale theorem. arXiv:2203.04485.

Scott, A. J. and T. M. F. Smith (1974). Analysis of repeated surveys using time series methods. *Journal of the American Statistical Association 69*(347), 674–678. https://doi.org/10.1080/01621459.1974.10480187.

Self, S. G. and K.-Y. Liang (1987). Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association 82*, 605–610.

Shin, J., A. Ramdas, and A. Rinaldo (2023). E-detectors: A nonparametric framework for sequential change detection. *The New England Journal of Statistics in Data Science 2*(2), 229–260. https://doi.org/10.51387/23-NEJSDS51.

Shin, J., A. Ramdas, and A. Rinaldo (2024). E-detectors: a nonparametric framework for sequential change detection. *New England Journal of Statistics in Data Science 2*, 229–260.

Siegmund, D. and E. S. Venkatraman (1995). Using the generalized likelihood ratio statistic for sequential detection of a change-point. *Annals of Statistics 23*, 255–271.

U.S. Census Bureau (2025). Business Trends and Outlook Survey: Methodology. U.S. Census Bureau, Washington, DC. https://www2.census.gov/data/experimental-data-products/business-trends-and-outlook-survey/methodology.pdf.

Vogelsang, T. J. (1998). Trend function hypothesis testing in the presence of serial correlation. *Econometrica 66*, 123–148.

Vovk, V. and R. Wang (2021). E-values: Calibration, combination and applications. *Annals of Statistics 49*(3), 1736–1754. https://doi.org/10.1214/20-AOS2020.

Wasserman, L., A. Ramdas, and S. Balakrishnan (2020). Universal inference. *Proceedings of the National Academy of Sciences 117*(29), 16880–16890. https://doi.org/10.1073/pnas.1922664117.

A Proofs

Proof of Theorem 2. Fix $\theta \in \Theta_0$, c and κ^2. Let $\mathcal{G}_n = \sigma(u_n)$ with $u_n = R_n/|R_n|$ and $R_n = M_n y_{1:n}$. Since the column space of X_{n+1} restricted to the first n rows is that of X_n (a column for a new regime is zero on earlier rows), $R_n = A_n R_{n+1}$ for a linear map A_n, hence $u_n = A_n u_{n+1}/|A_n u_{n+1}|$ is a function of u_{n+1} and $(\mathcal{G}_n)$ is a filtration. The law of R_n is $\mathcal{N}(0, \kappa^2 M_n \Sigma_\theta M_n^\top)$, so u_n has the angular central Gaussian law with shape $\Sigma_R = M_n \Sigma_\theta M_n^\top$, whose density with respect to the uniform measure on the sphere of the residual space is $C_\nu \, |\Sigma_R|^{-1/2} (u^\top \Sigma_R^+ u)^{-\nu/2}$; it does not depend on κ^2 or c. With $R = |R| u$, $u^\top \Sigma_R^+ u \, |R|^2 = y^\top P_\theta y$ and the pseudo-determinant of Σ_R is

$|\Sigma_\theta||X_n^\top\Sigma_\theta^{-1}X_n|/|X_n^\top X_n|$, which gives the expression of Section 4 up to factors common to all θ and components. The laws of u_n under a consistent family of Gaussians are consistent in n (the law of u_n is the image of the law of u_{n+1}), so $\Lambda_n = Q_n(u_n)/p_\theta(u_n)$ is the Radon–Nikodym derivative of Q with respect to P_θ on $\mathcal{G}_n$ and a nonnegative P_θ-martingale with $\Lambda_0 = 1$; $Q_n \ll p_\theta$ because every angular central Gaussian with positive-definite shape has a strictly positive density. Since $\theta \in \Theta_0$, $\sup_{\theta'} p_{\theta'}(u_n) \geq p_\theta(u_n)$, so $0 \leq E_n \leq \Lambda_n$. By Ville's inequality $\mathbb{P}(\sup_n \Lambda_n \geq 1/\alpha) \leq \alpha$, which contains $\{\exists n : E_n \geq 1/\alpha\}$. For the profile-scale variant replace u_n by R_n and Θ_0 by the set of pairs (κ^2, r). The Corollary follows because each listed covariance family is consistent in n, and a union of classes is a larger Θ_0. □

Derivation of (2). Integrating $\delta \sim \mathcal{N}(0, \kappa^2\tau^2)$ out of $\mathcal{N}(\delta Mz, \kappa^2 M\Sigma_{r_g}M^\top)$ gives $\mathcal{N}(0, \kappa^2 M(\Sigma_{r_g} + \tau^2 zz^\top)M^\top)$. By the matrix determinant lemma the pseudo-determinant is multiplied by $1+\tau^2 b$, with $b = z^\top P_{r_g} z$, and by the Sherman–Morrison formula the quadratic form is $y^\top P_{r_g} y - \tau^2 a^2/(1+\tau^2 b)$ with $a = z^\top P_{r_g} y$. Inserting both into the angular density gives (2). □

Proof of Proposition 2. $E_n = Q_n/\max_r p_r \geq w_c q_c/\max_r p_r = w_c\,\Lambda_{c,n}\,p_{r_g}/\max_r p_r$. Take logarithms. □

Proof of Theorem 3. (i) For any θ with $\mathrm{KL}(Q\|P_\theta) < \infty$ and $E \geq 0$, $\mathbb{E}_Q \log E = \mathrm{KL}(Q\|P_\theta) + \mathbb{E}_Q \log(E\, dP_\theta/dQ) \leq \mathrm{KL}(Q\|P_\theta) + \log \mathbb{E}_\theta E \leq \mathrm{KL}(Q\|P_\theta)$ by Jensen's inequality and $\mathbb{E}_\theta E \leq 1$; take the infimum over θ. (ii) The data-processing inequality for the indicator of A gives $\mathrm{kl}(Q(A)\|P_\theta(A)) \leq \mathrm{KL}(Q\|P_\theta)$. If $Q(A) > \alpha \geq P_\theta(A)$ then $\mathrm{kl}(Q(A)\|x)$ is decreasing in $x \leq Q(A)$, so $\mathrm{kl}(Q(A)\|\alpha) \leq \mathrm{kl}(Q(A)\|P_\theta(A)) \leq \mathrm{KL}(Q\|P_\theta)$ for every θ; hence $\mathrm{kl}(Q(A)\|\alpha) \leq K$ and $Q(A) \leq \beta^*$. (iii) On the ν-dimensional residual space, $\mathrm{KL}(\mathcal{N}(m, S_g)\|\mathcal{N}(0, \kappa^2 S_r)) = \frac{1}{2}[\mathrm{tr}(S_r^{-1}S_g)/\kappa^2 + m^\top S_r^{-1} m/\kappa^2 - \nu + \nu \log\kappa^2 + \log|S_r| - \log|S_g|]$ with $m = \delta Mz$, $S_g = \kappa_g^2 M\Sigma_{r_g}M^\top$ and $S_r = M\Sigma_r M^\top$. Minimizing over κ^2 gives $\kappa^2 = c_r/\nu$ with $c_r = \mathrm{tr}(S_r^{-1}S_g) + m^\top S_r^{-1} m$ and the value in the statement, where $m^\top S_r^{-1} m = \delta^2 z^\top P_r z$. At $r = r_g$, $c_r = \nu\kappa_g^2 + \delta^2 z^\top P_{r_g} z$ and the value is $\frac{1}{2}\nu\log(1 + \delta^2 b_g/\nu) \leq \frac{1}{2}\delta^2 b_g$. For the scale-exact monitor the data are the direction u_n, a function of y, and the data-processing inequality gives $\mathrm{KL}(Q_u\|P_{\theta,u}) \leq \mathrm{KL}(Q\|P_\theta)$, so that (i) and (ii) hold with K computed on the full data. □

Proof of Theorem 1. Write $S_i = \sum_{j \geq i} a_j$. Since $a \perp (1, j)$ the linear trend and L_1 do not enter $a^\top y = a^\top L + a^\top e$, and with $L_j = L_1 + \sum_{i=2}^{j} \eta_i$ we have $a^\top L = \sum_{i=2}^{n} S_i \eta_i$. By independence, $\mathrm{Var}(a^\top y) = p\sum_{i \geq 2} S_i^2 + \sigma^2\|a\|^2$, and $\sum_{i \geq 2} S_i^2 = a^\top W a$ up to terms that vanish when $a \perp 1$. Hence $\mathrm{Var}(Z_a) = 1 + (p/\sigma^2)\, a^\top W a/\|a\|^2$. Since ψ is piecewise continuous, the residual $a_j = \tilde\psi(j/n) + O(n^{-1})$ uniformly off the $O(1)$ discontinuities, so $S_i = n\,\Psi(i/n) + O(1)$ and $\sum_i S_i^2 = n^3 \int_0^1 \Psi^2\,(1 + o(1))$, while $\|a\|^2 = n\int_0^1 \tilde\psi^2\,(1 + o(1))$. This gives the stated form. The constant $c(\psi)$ is positive: $\int \tilde\psi^2 > 0$ because ψ is not linear, and $\Psi \equiv 0$ would force $\tilde\psi = 0$ almost everywhere. For $\psi = x^2$, $\tilde\psi = (x - \frac{1}{2})^2 - \frac{1}{12}$, $\int \tilde\psi^2 = 1/180$ and $\Psi(x) = \int_x^1 \tilde\psi$ has $\int \Psi^2 = 1/7560$, so $c = 180/7560 = 1/42$.

For the e-process, Z_a is Gaussian with mean zero and variance $1 + \lambda_n \to \infty$, so $\mathbb{P}(|Z_a| > z_n) \to 1$ for any $z_n = o(\sqrt{\lambda_n})$. With $A^2/B = Z_a^2$, $\log E_n(a) = -\frac{1}{2}\log(1 + \tau^2 B) + \frac{1}{2} Z_a^2\, \tau^2 B/(1 + \tau^2 B)$ and $B = \|a\|^2/\sigma^2 = O(n)$, which exceeds $\log(2/\alpha)$ once $Z_a^2 \geq 2\log(2/\alpha) + \log(1 + \tau^2 B) + O(1) = O(\log n)$; this has probability tending to one since Z_a^2 is of order n^2. The half-normal mixture is bounded below by half the normal-mixture value times a constant factor, and the argument is unchanged. □

Sketch for Proposition 1. Let $\mathrm{RSS}_n^{(0)}$ be the residual sum of squares of y on $(1, j)$ and $\rho_n = (a^\top y)^2/(\|a\|^2 \,\mathrm{RSS}_n^{(0)})$ with a the curvature contrast. The profile likelihood ratio of the quadratic model against the linear one is $(1-\rho_n)^{-\nu/2}$ with $\nu = n-2$, and a Laplace approximation of the mixture over the curvature gives $\log E_n^\sigma = \frac{\nu}{2}\log\{1/(1-\rho_n)\} + O_P(\log n)$. By Donsker's theorem, $(np)^{-1/2} L_{\lfloor nx \rfloor}$ converges to a Brownian motion, and ρ_n converges in distribution to the squared cosine between the detrended Brownian path and the quadratic, which is positive almost surely. On the event $\rho_n > \varepsilon$, whose probability tends to $\mathbb{P}(\rho_\infty > \varepsilon)$, $\log E_n^\sigma$ is of order n; letting $\varepsilon \to 0$ gives the claim. We present this as a sketch; the full argument requires uniform control of the Laplace remainder.

Computation. The e-process is computed on the log scale. For each wave n the matrices $P_r = \Sigma_r^{-1} - \Sigma_r^{-1} X (X^\top \Sigma_r^{-1} X)^{-1} X^\top \Sigma_r^{-1}$, $\Sigma_r = D + rW$, are precomputed for the roughness grid, and the quadratic forms $y^\top P_r y$, $z^\top P_r y$ and $z^\top P_r z$ give E_n in closed form from (2).